ZHANHAO (NEO) ZHANG

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EDUCATION

Cornell University, Ithaca, NY Expected May 2027
Ph.D. in Operations Research | Advisors: Jim Dai, Manxi Wu

Columbia University, New York, NY Feb 2022
M.A. in Statistics

University of Washington, Seattle, WA Jun 2020
B.S. in Computer Science; B.S. in Statistics

INDUSTRY EXPERIENCE

Meta, Seattle, WA May 2026 - Aug 2026
MACHINE LEARNING INTERN

- Built an LLM-based advertiser simulation framework to model budget-allocation decisions without direct competitor data, using advertiser trajectories and personas to capture responses to Meta's relative positioning.
- Developed a counterfactual evaluation framework to estimate advertiser responses to alternative signal strategies, overcoming the limitation that historical data capture only realized decisions rather than outcomes under unobserved alternatives.
- Designed an interpretable LLM-augmented reinforcement learning method over fixed-structure decision-tree policies, enabling policy optimization while satisfying interpretability and low-latency serving requirements.
- Redesigned the simulation pipeline with parallelization and caching, reducing end-to-end runtime by ~6x and model-serving cost by ~20x; core trajectory-generation components were adopted by the broader team.

Akamai Technologies, Boston, MA Jun 2025 - Aug 2025
RESEARCH INTERN, MAPPING

- Modeled content-delivery-network traffic using stochastic methods and built a simulation pipeline to evaluate machine utilization under DNSP load-balancing policies.
- Designed and evaluated load-balancing policies to improve tail latency for end users.

Aetna at CVS Health, New York, NY Jun 2021 - Jul 2022
DATA SCIENTIST

- Built a distributed Monte Carlo simulation pipeline for member-disenrollment estimation, completing 100K+ simulations on datasets with millions of rows within one hour through parallelization and vectorization.
- Developed optimization and machine-learning methods for campaign experimental design, risk-cohort identification, and member churn analysis using quadratic programming, Dask, Hive, and supervised learning.

SELECTED RESEARCH

Atomic Proximal Policy Optimization for Electric Robo-Taxi Dispatch and Charger Allocation *Minor Revision, Transportation Science*
Jim Dai, Manxi Wu, **Zhanhao Zhang**

Optimal Batched Scheduling of Stochastic Processing Networks Using Atomic Action Decomposition *Major Revision, Management Science; Revision Submitted*
Jim Dai, Manxi Wu, **Zhanhao Zhang**

Deep Learning Method for Stationary Distribution of Reflected Brownian Motion *Submitted, Stochastic Systems*
Jim Dai, **Zhanhao Zhang**

Generative Market Equilibrium Models with Stable Adversarial Learning via Reinforcement Link *Major Revision, SIAM Journal on Financial Mathematics; Revision Submitted*
Anastasis Kratsios, Xiaofei Shi, Qiang Sun, **Zhanhao Zhang**

Mixed-Type Courier Dispatch for Online Food Delivery Platforms *Major Revision, Transportation Science; Revision Submitted*
Junlin Chen, Manxi Wu, Chiwei Yan, **Zhanhao Zhang**

HONORS & TEACHING

Outstanding Teaching Assistant Award, Cornell ORIE | 2026
INFORMS TSL Data-Driven Research Challenge - Finalist | May 2025
Cornell Fellowship, Cornell University | Jan 2023 - May 2023
Instructor, Big Data Technologies, Cornell University | Spring 2024, 2025, 2026

TECHNICAL SKILLS

Methods: Reinforcement Learning, LLM Agents, Optimization, Stochastic Modeling, Simulation, Parallel/Distributed Computing
Languages: Python, SQL, C/C++, R, Bash
ML / Scientific Computing: PyTorch, NumPy, SciPy, scikit-learn, Pandas
Data / Systems: PySpark, Hive, Joblib, Git